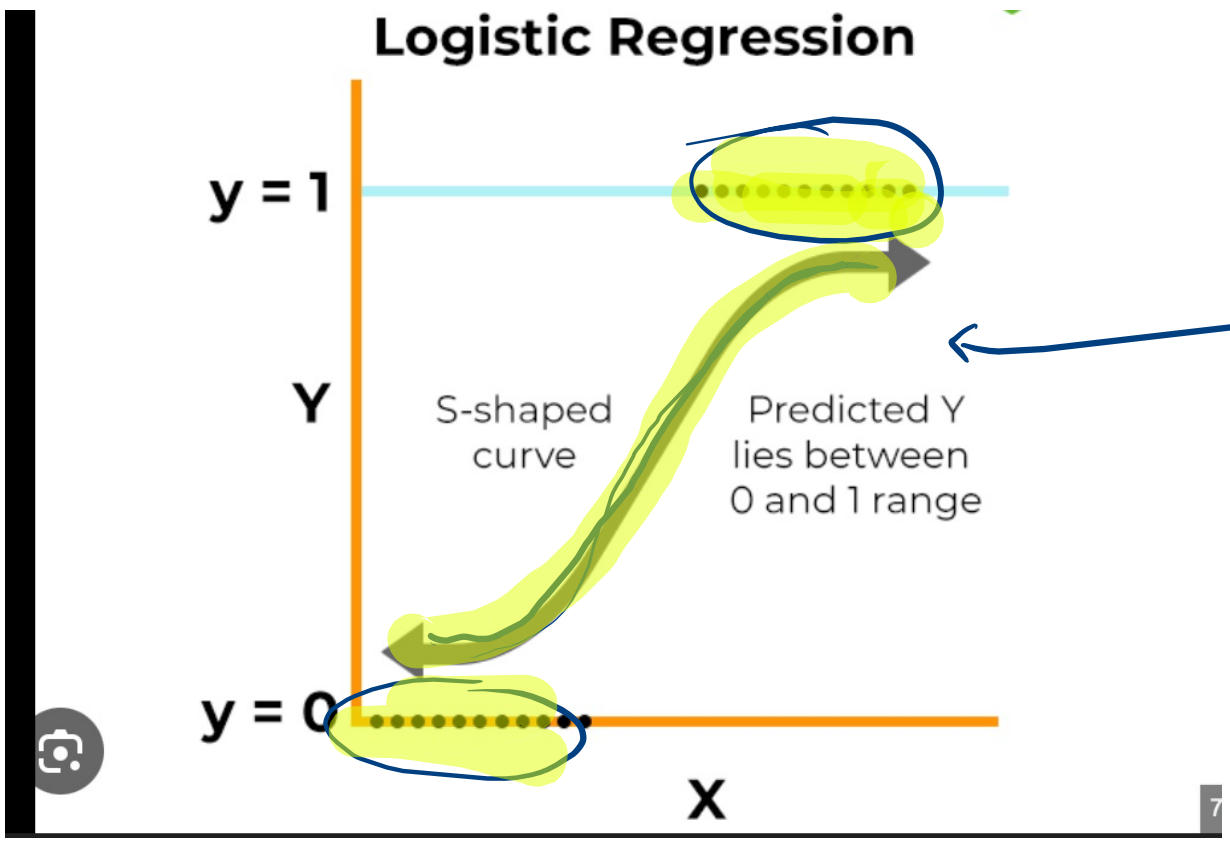
Used for classification problem. ← Logistic Regression

Output → 0 or 1 (yes/no, Pass/fail)



(S shaped curve)

Sigmoid curve

Output → Probability (0 to 1)

Logistic Regression is used to predict categorical outcomes (0 or 1) using probability.

Real problem → Predict → will student pass or fail?

Student Hours	Pass (1/0)
1	0
2	0
3	0
4	1
5	1

Output = $\frac{3}{5}$ → Pass

Output = $\frac{2}{5}$ → fail

Sigmoid function ⇒ $p = \frac{1}{1 + e^{-z}}$

Sigmoid is a mathematical function that converts any value into a Range b/w 0 and 1.

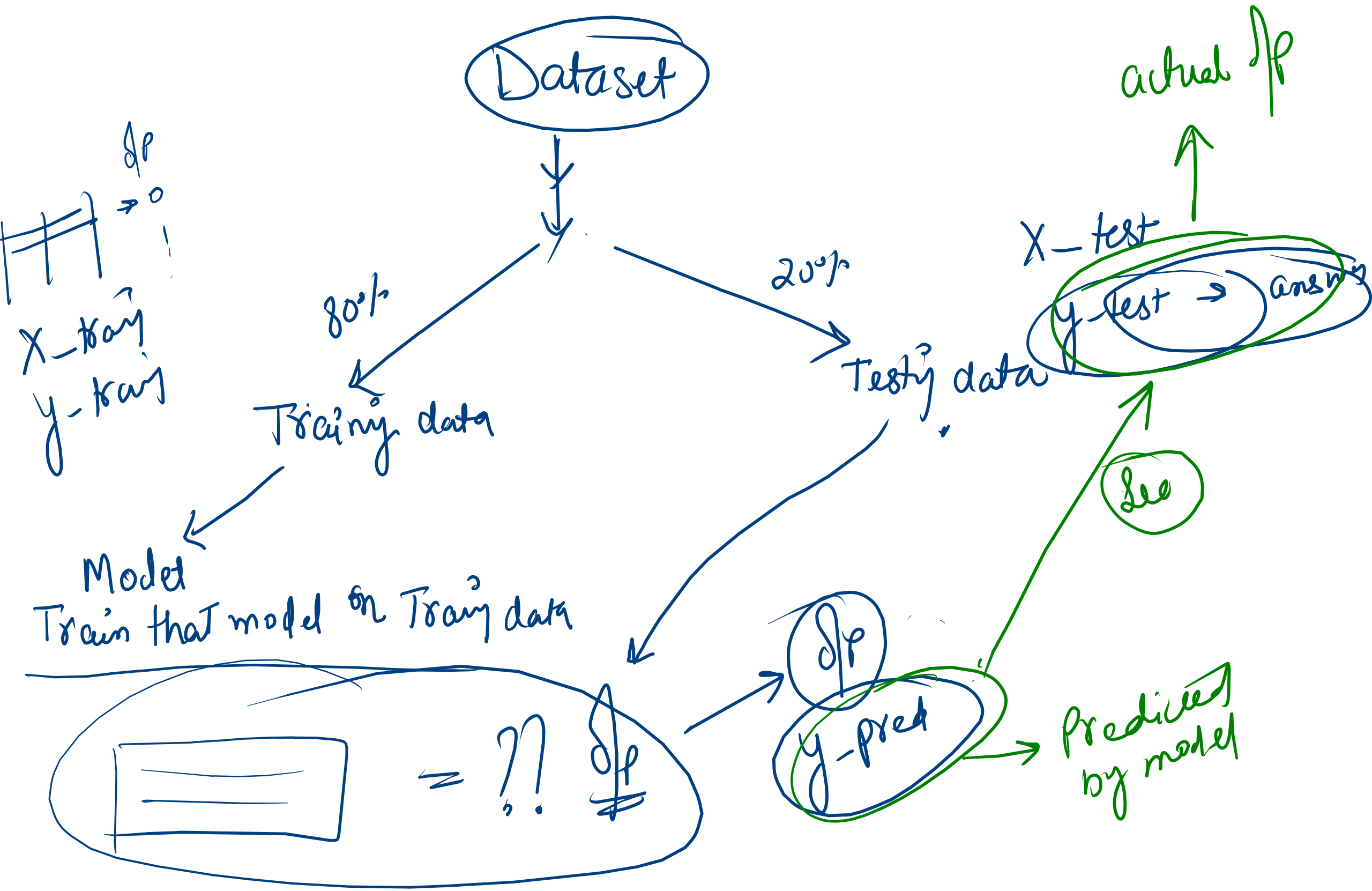
← Sigmoid function

$$\sigma = \frac{1}{1 + e^{-z}}$$

mathematical constant (2.718)

output of linear equation.

$x = my + b$


Dataset
80%
20%
Training data
Testing data
X_train
y_train
o/p
X_test
y_test
answer
actual o/p
Model
Train that model on Train data
= ?? o/p
o/p
y_pred
Predicted by model

KNN

(K-Nearest Neighbour)

KNN is a non parametric, lazy learning algorithm that predicts output based on closest data points in the dataset.

No fixed formula

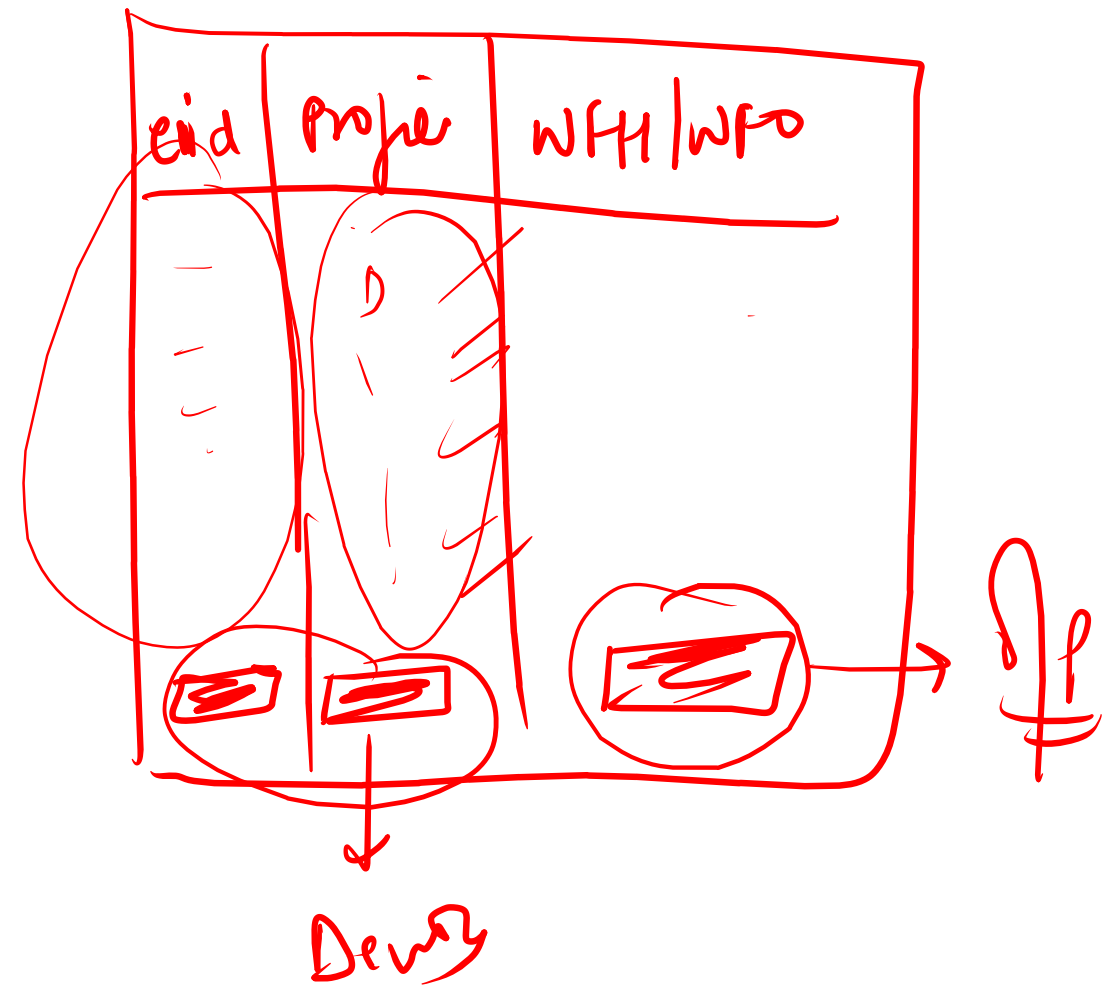
(Unlike linear / logistic Regression)

Lazy learning

No actual 'training'

Model just stores the data.

Office → Google



Study Hours	Result
1	Fail
2	Fail
3	Pass
4	Pass

Pick k nearest

[3, 2, 1]

3 → Pass, 2 → Fail, 1 → Fail

Predict = Fail

2.5 hours

Choose

$x_1 = K_1 = 3$ (Nearest value)

$x_2 = K_2 = 2$ (second nearest value)

Calculate distance $= \sqrt{(x_1 - x_2)^2}$

$= \sqrt{(3-2)^2}$

$= 1$

Age, Income, Season → Tshirt

Jeans

20, $1000, Summer → Tshirt

Jacket

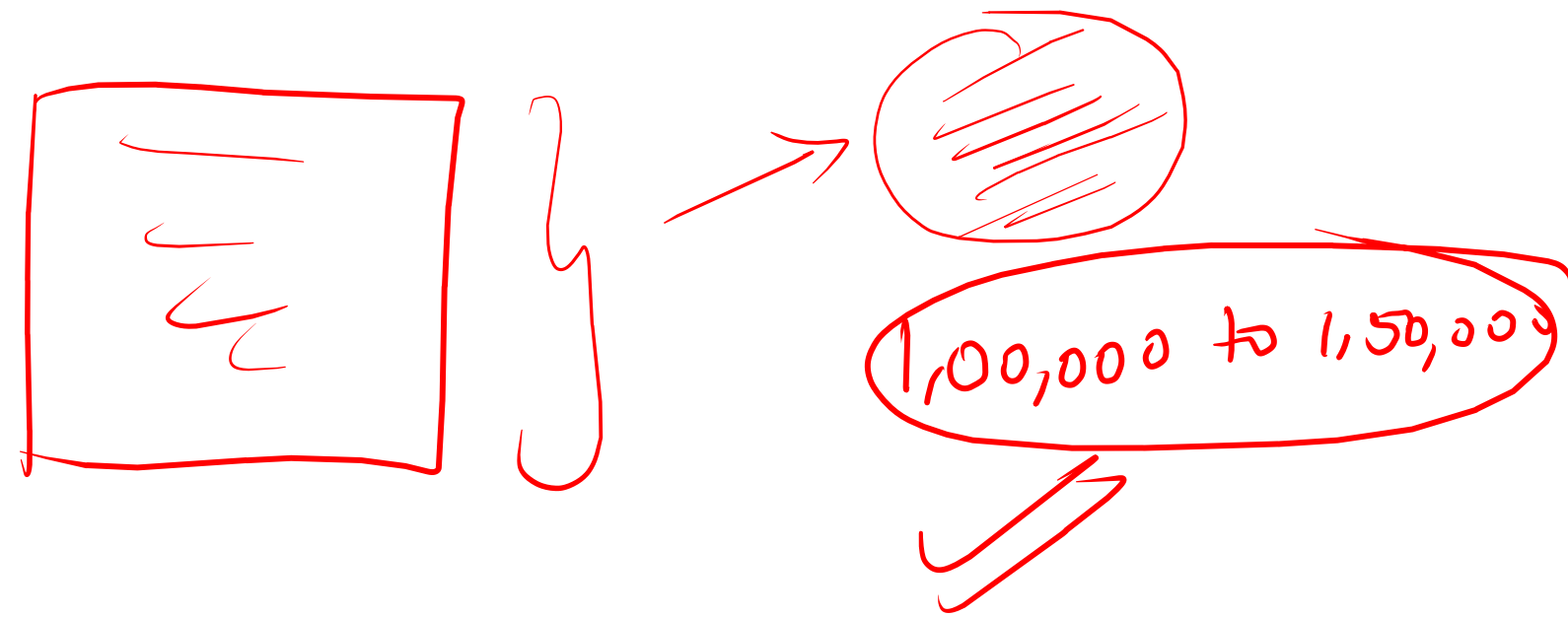
Overfitting

Overfitting happens when a model learns too much from Training data, including noise.

Income = 50 → jeans

Underfitting

Underfitting happens when the model is too simple to learn the patterns.



Bias vs variance

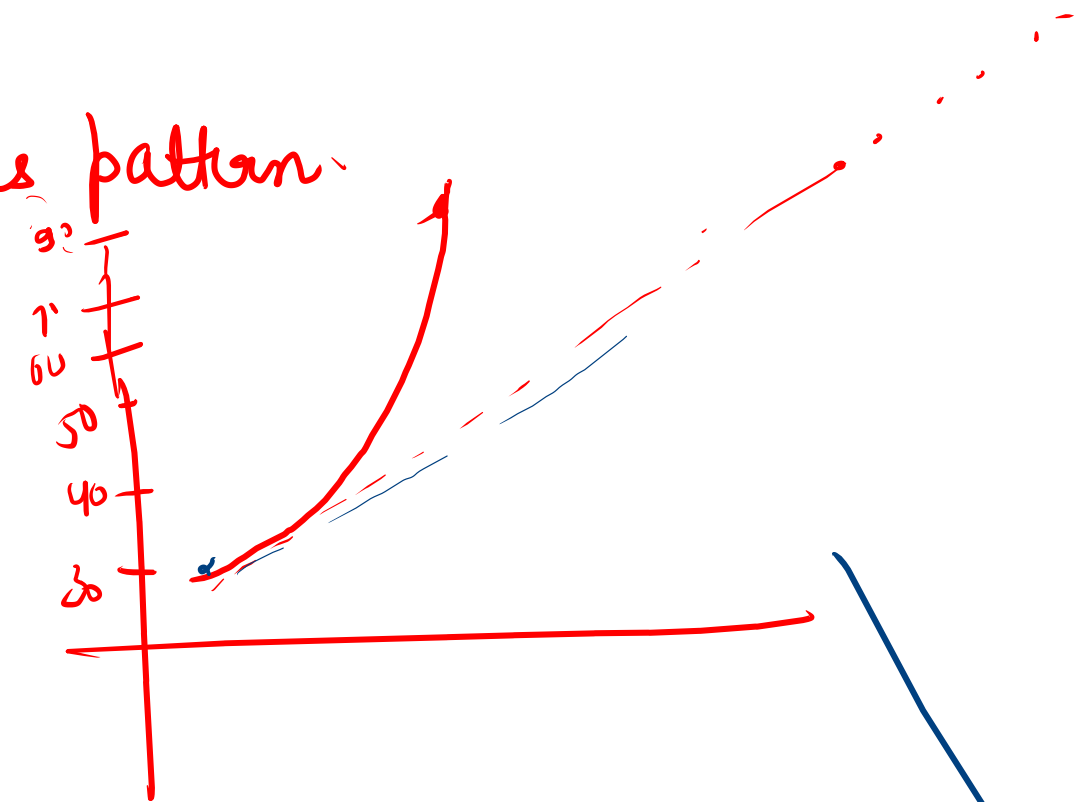
Bias → model is too simple

What happens → model cannot learn properly

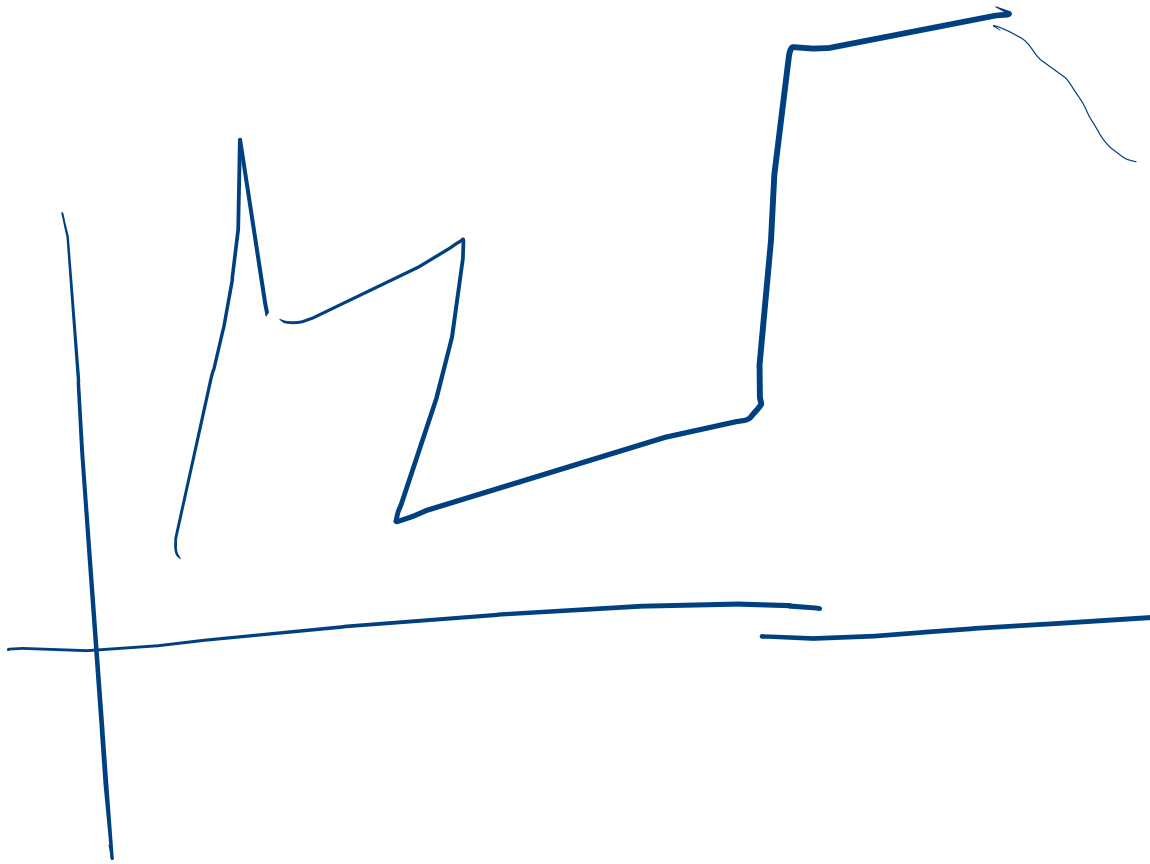
→ misses pattern.

Data :-

Hours	Marks
1	30
2	40
3	60
4	90



Wrong predictions
Poor performance



Variance

Model is too complex

What happens ?

→ Model learn every detail + noise.

→ Memorize data

Model creates very complicated curve — passes through every point exactly.

Result → Perfect on Training data
fails on new data

⇒ High variance

Bias → Not learning enough

Variance → learning too much.

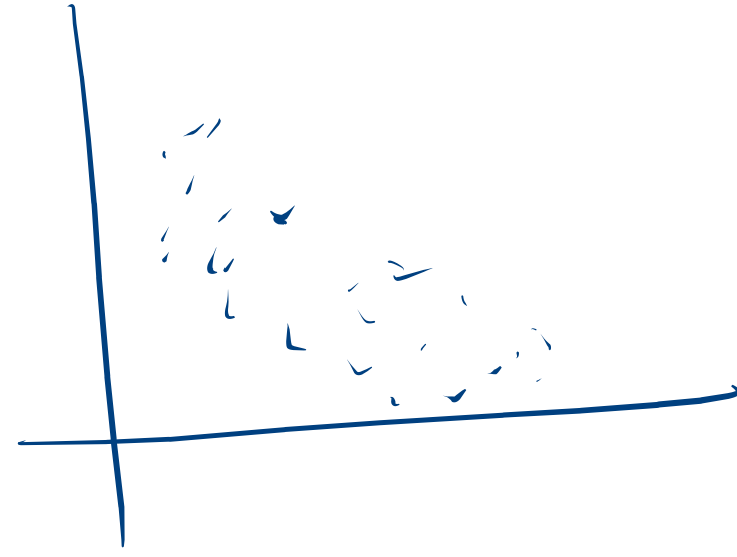
SMOTE → Technique used to balance imbalanced dataset by creating synthentic data points.

Imbalanced

Class	Count
TShirt	90
Jacket	10

⇒ Always predict TShirt

SMOTE works :-



1. Pick a minority class points
2. find its nearest neighbors
3. Create new point b/w them

So the new data look realistic